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paired comparison and rank order data

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Building upon Bradley-Terry and Plackett-Luce: some methods for modeling paired comparison and rank order data

A DISSERTATION PRESENTED

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Building upon Bradley-Terry and Plackett-Luce: some methods for modeling paired comparison and rank order data

ABSTRACT

A common model used for analyzing paired comparison data is the Bradley-Terry model. An implicit assumption embedded within the Bradley-Terry framework is stochastic transitivity. This implies that if A is preferred to B, and B is preferred to C, then A must be preferred to C. Chapter 1 proposes two Bayesian hierarchical paired comparison models that extend the Bradley-Terry model by introducing a stability parameter for each player. This additional dimension on player characteristics permits the violation of stochastic transitivity among players. This can improve paired comparison inferences fit and out-of-sample prediction accuracy when there is evidence of violation of stochastic transitivity in the data. Chapter 2 extends the model developed in Chapter 1 by developing a rating system that characterizes athletes through strength and stability parameters. We find that the new rating system is capable of identifying athletes with high level of instability.

Chapter 3 evaluates a common practice of analyzing rank order data (e.g., from multi-competitor races) by converting them into paired comparison data and using Bradley-Terry models to make inferences and predictions. We establish this practice within existing statistical frameworks, assuming the rank order data can be analyzed via the conventionally used Plackett-Luce model, and prescribe the necessary adjustments to the covariance matrix of the Bradley-Terry estimators. We also describe the loss of efficiency in doing so. The second part of Chapter 3 draws on the formulation of rank order data as paired comparisons to propose a new model diagnostic tool for the Plackett-Luce model. In a simulation study, this new model diagnostic has good power in detecting departures from Plackett-Luce models.

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PREVIEW

I DEDICATE THIS THESIS TO MY PARENTS AND MY GRANDPARENTS.

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0

Introduction

The field of quantitative analysis in sports has become substantially more established over the last two decades. One intrinsic source of data from sports event is the observed outcome of sports competitions. We can divide these sports into two broad categories: head-to-head sports and multi-competitor sports.

In head-to-head sports, exactly two players or teams directly face each other during the game. Examples of head-to-head sports include team sports such as baseball, soccer and basketball, individual

sports such as tennis and badminton, mind sports such as chess and e-sports such as League of Legends and Counter-Strike. Outcomes of these games can be reported as which player (team) won the game. The results of head-to-head games are examples of paired comparison data. In some sports such as tennis and volleyball, the outcome of the game is always binary. The rules of these sports dictates that each game must finish with an unequivocal winner. In other sports such as chess, games can finish with a winner or in a tie.

In multi-competitor sports, players (or teams) perform a specific task, and are compared against each other based on some performance measure. For example in sports such as swimming, running and skiing, athletes' performance are measured by the time taken to finish a particular distance. In sports such as diving, ice skating and gymnastics, athletes are judged by a common panel by how well they performed a routine. These competitions can be summarized by listing players' performance from best to worst. Such records are examples of rank order data.

A common model used for analyzing paired comparison data is the Bradley-Terry model (Bradley and Terry, 1952), while rank order data is often modeled using the Plackett-Luce model (Plackett, 1975; Luce, 1959). Both models represent individual athletes' average skills using a numerical value, and prescribe the distribution of outcomes of the game as a function of these athletes' average skills. Using these models, we can obtain estimates of athletes' relative strength and make predictions regarding unobserved or future outcomes.

The first two chapters of this dissertation focus on the analysis of paired comparison data. An implicit assumption embedded within the Bradley-Terry framework is stochastic transitivity. In the context of modeling athletes' abilities, stochastic transitivity implies that if player A is stronger than player B, and player B is stronger than player C, then player A must be stronger than player C. This can be a restrictive assumption. In Chapter 1, to accommodate the violation of stochastic transitivity, we propose two variants of a Bayesian hierarchical paired comparison model. The hierarchical models we propose extends the Bradley-Terry model by introducing a stability parameter for each player. This

additional dimension on player characteristics permits the violation of stochastic transitivity among players. Our models embed the Bradley-Terry model as a limiting case when the stability parameters are exactly 0. Using simulation and empirical examples, we demonstrate that our hierarchical models can outperform the Bradley-Terry model when the data deviates from Bradley-Terry assumptions.

Chapter 2 extends the concept of characterizing athletes using average strength and stability. We propose a dynamic paired comparison model with time-varying strength and stability parameters. Paralleling the development of the Glicko (Glickman, 1999) and Glicko-2 (Glickman, 2001) rating systems, we adapt the dynamic paired comparison model into a new rating system. Given new observations from a fixed time period, the rating system updates the estimates of players' strength and stability. We demonstrate this rating system using data from professional Go.

The first two chapters introduce methodologies for a new modeling framework for outcomes from head-to-head sports. They accommodate the natural phenomenon of stochastic intransitivity. The new framework quantifies each athletes' ability through strength and stability. This introduces additional insights on athletes that is not present in Bradley-Terry models and can improve the predictive capabilities of the models. The methods introduced in the first two chapters can also be extended to analyses of paired comparison data in contexts other than sports. In Chapter 1, we demonstrate the application of our model to a citation data set.

The third chapter of this dissertation examines modeling of rank order data. We are aware of instances where practitioners in the field of sports analytics choose to analyze rank order data by converting the data into sets of paired comparisons. Instead of using the Plackett-Luce model, the Bradley-Terry model is applied instead to the formed paired comparisons. We would like to help these practitioners appreciate what they are doing from a statistical perspective. The first part of Chapter 3 establishes the validity of this approach. We also prescribe the necessary adjustments to the covariance matrix of the Bradley-Terry estimators, and describe the loss of efficiency in doing so. The second part of Chapter 3 draws on the formulation of rank order data as paired comparisons to propose a

new model diagnostic tool for the Plackett-Luce model. We demonstrate the behavior of this new model diagnostic through simulation studies. Using a simulation study, we show that this model diagnostic can detect departures from Plackett-Luce model and has good power in doing so. In short, the third chapter affirms an observed practice among sports analytics professionals in dealing with the analysis of rank order data. The chapter also introduces a new tool that can be used for checking the appropriateness of the Plackett-Luce model when modeling outcomes from multi-competitor sports.

PREVIEW

1

Hierarchical Models for Modeling Violation of Stochastic Transitivity in Paired Comparison Data

1.1 INTRODUCTION

Paired comparison data arise out of observing the direct competition between two players, objects, or options. Common examples can be observed in head-to-head sports competitions, with each game involving two players facing off against each other. Examples include common sports like soccer, baseball, basketball and chess. Many other problems can also be analyzed under this framework. For example, analysis of observed data on consumer preferences between two brands can be seen as a competition between these two brands.

A classic and popular family of models for such problems is the family of linear paired comparison models (David, 1988). This family of models, and extensions based on these models, are extensively used in analyzing head-to-head sports competitions. It has also been extended to other context such as analyzing citation networks (Stigler, 1994; Varin et al., 2016), understanding decision-making in starting a doctoral program (Dittrich et al., 2004) and modeling organisms' preferences in Ecology (Müller et al., 2015; De'ath and Moran, 1998; Larsen et al., 2008). The framework of paired comparisons and the application of linear paired comparison models can be found in a wide range of academic disciplines.

An implicit assumption of this family of linear paired comparison models is stochastic transitivity. We describe stochastic transitivity in the context paired competition among sports players. For any arbitrary player triplet A, B and C, if player A is stronger than player B, and player B is stronger than player C, stochastic transitivity implies that player A must be stronger than player C. We argue that this might be a restrictive assumption, and would like to contribute a way of allowing for violations of stochastic transitivity while modeling paired comparisons.

Linear paired comparison models characterize each player by a strength parameter. The main idea behind our work is to introduce an additional parameter at the player level which we refer to as the stability parameter. With this additional parameter, players can differ in strength and/or stability.

This extension allows violations of stochastic transitivity to exist. A simple analogy can be compared to the problem of stochastic dominance between random variables that are Normally distributed. If we fix the variance parameter, then the random variable with a higher mean parameter must stochastically dominate the other. However, when we allow the variance parameter to change, the stochastic dominance relationship between variables becomes inconclusive.

The remainder of this chapter is organized as follows. Section 1.2 reviews linear paired comparison models and current work on accommodating intransitivity. To account for intransitivity, we introduce two variants of a Bayesian hierarchical paired comparison model in Sections 1.3 and 1.4. For each of our models, we discuss interpretations of model parameters and describe computational details for drawing posterior samples. We demonstrate the new models in Section 1.5 and 1.6 using simulations and empirical data, respectively. In Section 1.5, the new models are compared against the Bradley-Terry model by evaluating the models' ability to recover true parameters based on simulated data. In Section 1.6, the new models are compared against the Bradley-Terry model by conducting out-of-sample predictions. The chapter concludes in Section 1.7.

1.2 STOCHASTIC TRANSITIVITY AND PAIRED COMPARISON MODELS

For simplicity, we assume that a comparison between two players can only result in one of two outcomes, a win or a loss. With a total of N players, given n_{ij} games played between player i and player j , we observe Y_{ij} cases where player i wins, for $i = 1, \dots, N - 1$; $j = i + 1, \dots, N$. The choice of $i < j$ is purely arbitrary for accounting purposes to avoid double counting player pairs. We define $Y_{ij} = n_{ij} - Y_{ji}$ for $i > j$ where needed.

We assume that every outcome resulting from games played between player i and player j are independently and identically distributed, and games across different player pairs are independent.

We can write this as

$$Y_{ij} \sim \text{Binom}(n_{ij}, p_{ij})$$

independently.

1.2.1 LINEAR PAIRED COMPARISON MODELS

The family of linear paired comparison models assumes that p_{ij} , the probability of player i defeating player j , can be described as

$$p_{ij} = F(\theta_i - \theta_j)$$

where F is the cumulative distribution function (CDF) of a continuous random variable with a density that is symmetric around 0. The zero-symmetric condition ensures that $p_{ij} = 1 - p_{ji}$. Setting F as the CDF of the standard Logistic distribution gives the classic Bradley-Terry model for paired comparison (Bradley and Terry, 1952), which can be expressed as

$$\text{logit}(p_{ij}) = \theta_i - \theta_j.$$

Another conventional choice, the CDF of the standard Normal distribution, gives the Thurstone-Mosteller model (Thurstone, 1927; Mosteller, 1951). We focus on the Bradley-Terry model in this chapter.

Frequentist inference on the Bradley-Terry model is typically done using maximum likelihood estimation. The problem of solving for the maximum likelihood estimator (MLE) of the Bradley-Terry model has been discussed by Hunter (2004). The Bradley-Terry model is also equivalent to a logistic regression with a special design matrix. Hence, the Bradley-Terry model can also be analyzed using standard software packages that perform logistic regression.

Consider paired comparisons among three distinct individuals i, j and k . Suppose that we know $p_{ij} \geq 0.5$ and $p_{jk} \geq 0.5$. Under weak stochastic transitivity, this implies $p_{ik} \geq 0.5$. Under strong stochastic transitivity, we have $p_{ik} \geq \max(p_{ij}, p_{jk})$ (Luce and Suppes, 1965). In this chapter we will focus on weak stochastic transitivity. Hereafter, stochastic intransitivity will refer to the existence of player triplets such that $p_{ij} \geq 0.5$, $p_{jk} \geq 0.5$ and $p_{ki} > 0.5$, that is, a violation of weak stochastic transitivity. Under the family of linear paired comparison models, we can see that

$$F^{-1}(p_{ik}) = F^{-1}(p_{ij}) + F^{-1}(p_{jk}).$$

When $p_{ij} \geq 0.5$ and $p_{jk} \geq 0.5$, the zero symmetry condition on cdf F guarantees that $F^{-1}(p_{ij}) \geq 0$ and $F^{-1}(p_{jk}) \geq 0$. Hence,

$$F^{-1}(p_{ik}) \geq \max(F^{-1}(p_{ij}), F^{-1}(p_{jk})) \geq 0.$$

Since F as a CDF is monotonically increasing, we see that the family of linear paired comparison models implicitly assumes strong and weak stochastic transitivity in all paired comparisons.

We argue that this is a strong assumption in thinking about paired comparison. For example, consider a student deciding to accept admission to colleges H, Y and P. Perhaps the student has a preference for college H when compared against Y because of their location, a preference for Y over P because of campus design but also a preference for P over H because of her parents' personal opinion on colleges P and H (this opinion may be non-existent on comparisons against Y). Here we arrived at a set of intransitive pairwise preferences such that $H \succ Y$, $Y \succ P$ and $P \succ H$, where " $\succ$ " means "is preferred to", because the pairwise decisions are not made based on the same considerations.

An empirical example of violation of stochastic transitivity is exhibited in data taken from an article on citations in Statistics journals (Varin et al., 2016). Stigler (1994) proposed to model the

log-odds of journal i being cited by j as the difference between a measure of each journal's academic strength in influencing other journals. This set-up for the analysis of citations across journals can be considered a paired comparison problem. Conditioning on number of citations exchanged between two journals, a citation of journal i by journal j can be viewed as a win for journal i in its ability to academically influence journal j . Table 1.1 gives the empirical probability of journal i being cited by journal j , for four selected Statistics journals. From the table, we can see that $\hat{p}_{\text{CSDA}, \text{JMA}} \geq 0.5$ and $\hat{p}_{\text{JMA}, \text{StMed}} \geq 0.5$, but $\hat{p}_{\text{CSDA}, \text{StMed}}$ is only 0.34. This illustrates violation of stochastic transitivity being observed.

	CSDA	JMA	JSPI	StMed
CSDA	-	0.54	0.36	0.34
JMA	0.46	-	0.47	0.58
JSPI	0.64	0.53	-	0.45
StMed	0.66	0.42	0.55	-

Table 1.1: Empirical probability of journals being cited by other journals from Varin et al. (2016).

Motivated by conceptual reasoning and empirical examples, we would like to examine how to extend linear paired comparison models, in particular, the Bradley-Terry model, to allow for violations of stochastic transitivity.

1.2.2 EXISTING WORK ON MODELLING INTRANSITIVITY

The Bradley-Terry model can be extended by assuming a set of $\binom{N}{2}$ parameters θ_{ij} and modeling the probability p_{ij} as

$$\text{logit}(p_{ij}) = \theta_i - \theta_j + \theta_{ij}. \quad (1.1)$$

Without any structure on θ_{ij} , the MLE of the p_{ij} for this model are the sample proportions $\frac{y_{ij}}{n_{ij}}$. To avoid overfitting, we can consider penalizing the log-likelihood using LASSO or ridge penalties. However, the θ_{ij} cannot be estimated if we do not have any observed data between two players. In other words, this model is unable to make inferences about player pairs who have not competed before without invoking additional assumptions.

Despite stochastic intransitivity being a such a natural phenomenon, the literature on modeling intransitivity is scarce. We recognize that one of the fundamental difficulties in developing a model is the prevalence of transitivity in mathematical operations. Early work by economist May (1954) discussed transitivity in the context of preferences and utility functions. He pointed out that a utility function built on a single dimension input must be transitive. Along these lines, several authors have attempted to allow for stochastic transitivity by extending scalar representations of player ability to multiple-parameter representations. The main tools in previous work has relied on vector operations such as measuring Euclidean distances and computing inner products.

Causeur and Husson (2005) proposed the model

$$\text{logit}(p_{ij}) = \sigma_{ij} \sqrt{(\lambda_{i,1} - \lambda_{j,1})^2 + (\lambda_{i,2} - \lambda_{j,2})^2},$$

where $\sigma_{ij} = -\sigma_{ji}$ is a sign parameter which takes value ± 1 . The vectors $\lambda_i = (\lambda_{i,1}, \lambda_{i,2})'$, $i = 1, \dots, N$ must also satisfy identifiability constraints

$$\sum_{i=1}^N \lambda_{i,1} = \sum_{i=1}^N \lambda_{i,2} = 0 \text{ and } \sum_{i=1}^2 \lambda_{i,1} \lambda_{i,2} = 0.$$

To summarize their approach, the Euclidean distance of the ability vectors λ_i and λ_j determines the gap between players i and j , while a separate sign parameter is used to indicate which player is stronger. This model does not provide structure to infer the sign parameter for players that have not played

against each other.

Chen and Joachims (2016) also used vector representation of player ability, where each player is represented by two d -dimensional vectors i_{blade} and i_{chest} . Then, the authors propose several ways to represent

$$\text{logit}(p_{ij}) = M(i, j)$$

such that violations of stochastic transitivity are allowed. Possible choices of the function $M(i, j)$ include

- $M(i, j) = ||j_{\text{blade}} - i_{\text{chest}}||_2^2 - ||i_{\text{blade}} - j_{\text{chest}}||_2^2$
- $M(i, j) = i_{\text{blade}} \cdot j_{\text{chest}} - j_{\text{blade}} \cdot i_{\text{chest}}$

The authors argued that these models with i_{blade} and i_{chest} can fully represent any structure in pairwise comparisons when the vectors are of sufficient length. The proof is based on vectors of length N and $N + 1$ for the inner product and Euclidean distance models respectively. A drawback of their approach is that their models were proposed as a pure prediction problem, and it is unclear how to extend their approach to account for statistical uncertainty and obtain parameter inferences.

1.2.3 MOVING AWAY FROM VECTOR REPRESENTATION

Rather than adopt an approach to modeling intransitivity which relies on operating on vector parameters, our proposed approach starts with a single scalar parameter θ_i to describe player i 's average ability, and then introduces a second parameter, δ_i , at the player level, so that player i is associated with a set of parameters (θ_i, δ_i) , in a way that will accommodate stochastic intransitivity. The role of δ_i will be distinct from θ_i . We give a brief overview of the broad idea below.

We choose to focus on representing θ_{ij} in Equation (1.1) in a structured approach. We separate

θ_{ij} as

$$\theta_{ij} = \Delta_{ij} - \Delta_{ji}$$

so that

$$\text{logit}(p_{ij}) = \theta_i - \theta_j + \Delta_{ij} - \Delta_{ji}.$$

We represent this deviation from a basic Bradley-Terry model into contributions by player i and player j respectively. Naively, this extension creates a model with unidentifiable parameters. However, we will be imposing a hierarchical structure on the parameters. We let $\Delta_{ij}, j = 1, \dots, N; j \neq i$, to share a common distribution G_i , where G_i is a parametric distribution characterized by the player specific parameter δ_i . The difference in distributions of Δ_i and Δ_k gives a second dimension in contrasting players i and k .

Note that when we move away from the Bradley-Terry model, we are giving up both stochastic transitivity and the fundamental assumptions of linear paired comparison models. Hence, our models can also be viewed as an alternative general approach to model paired comparison data which encompasses the Bradley-Terry model as a special case and allows violation of stochastic transitivity. In the following sections, we introduce two variants of this hierarchical model structure.

1.3 MODEL VERSION I

The first model we propose assumes the t distribution with degrees of freedom ν as our choice of G_i , the distribution on Δ_{ij} . This Bayesian hierarchical model can be seen as a continuous bridge between the Bradley-Terry model and a saturated model. The complete model specification with prior distribution is displayed in Table 1.2.

Data model

$$\begin{aligned} Y_{ij} &\sim \text{Binom}(n_{ij}, p_{ij}) \\ \text{logit}(p_{ij}) &= \theta_i - \theta_j + \Delta_{ij} - \Delta_{ji} \\ &\quad i, j = 1, \dots, N; j > i. \end{aligned}$$

Player level parameters

$$\begin{aligned} \boldsymbol{\theta} &\sim \mathcal{N}(0, \lambda^2 \mathbf{I}) \text{ s.t. } \sum_{i=1}^N \theta_i = 0 \\ \sigma_i &= k_i \lambda \\ k_i &\sim \text{Exp}(k_0) \\ &\quad i = 1, \dots, N. \end{aligned}$$

Player-pair parameters

$$\frac{\Delta_{ij}}{\sigma_i} \sim t_\nu \quad i, j = 1, \dots, N; j > i.$$

**Prior on
global parameters**

$$\begin{aligned} k_0 &\sim \Gamma(k_\alpha, k_\beta) \\ \lambda^2 &\sim \text{Inv-}\Gamma(\lambda_\alpha, \lambda_\beta) \end{aligned}$$

Table 1.2: Complete specification of Model 1.

We define $\Delta_{ii} = 0$ for convenience. We introduced player-level parameters k_i which can be interpreted as a measure of stability in performance of player i . We will demonstrate in Section 1.3.1.2 that the parameter k_i has the important role of controlling the prevalence of stochastic intransitivity in the model.

Similar to the Bradley-Terry model, we use θ_i to represent player i 's latent ability. Since the model likelihood is invariant to additive shifts in $\boldsymbol{\theta}$, we need to add a constraint. Instead of working with an arbitrary reference player by setting $\theta_{i_0} = 0$ for some arbitrary player i_0 , we choose to work with the identifying constraint $\sum_{i=1}^N \theta_i = 0$. This gives θ_i the interpretation of difference in player i 's average strength from the average player in the population. The constraint can be incorporated into the prior

distribution by re-writing it as

$$\theta_{1:(N-1)} \sim \mathcal{N} \left(0, \lambda^2 \left[\mathbf{I}_{N-1} - \frac{1}{N} \mathbf{1}_{N-1} \mathbf{1}_{N-1}^\top \right] \right),$$

and set $\theta_N = -\sum_{i=1}^N \theta_i$. Here, $\theta_{1:(N-1)}$ refers to the first $N - 1$ components of θ , $\mathbf{I}_{N-1}$ is the identity matrix of dimension $N - 1$ and $\mathbf{1}_{N-1}$ is the $(N - 1)$ -vector consisting of all 1s. The parameter λ controls the spread in player strength within the population. We will discuss in detail the influence of λ on the model in Section 1.3.1.1.

The log odds of player i defeating player j is characterized by the difference between θ_i and θ_j and the difference between Δ_{ij} and Δ_{ji} . The parameter Δ_{ij} can be interpreted as the average deviation of player i from player i 's average when facing opponent player j . Magnitudes of $\Delta_{i1}, \dots, \Delta_{iN}$ for player i are controlled by the variance σ_i^2 . A player with small σ_i has consistent performances when facing different opponents. A player with large σ_i can perform at vastly different strengths when facing different opponents. The standard deviation σ_i can be decomposed as product of λ and the stability parameter k_i . The distribution of the Δ_{ij} scaled by σ_i is assumed to be t -distributed with degrees of freedom ν . When ν is high, the distribution is approximately standard Normal. When ν is small, the distribution of $\frac{\Delta_{ij}}{k_i \lambda}$ has heavy tails. Under such model specifications, there is more room for Δ_{ij} to vary, and may risk over fitting to empirical probabilities. By default, we choose to work with the standard Normal distribution.

The Thurstone-Mosteller model can be described through a latent variable specification, with

$$U_i = \theta_i + \epsilon_i$$

where $\epsilon_i \sim \mathcal{N}(0, \tilde{\sigma}_i^2)$ for each player i , and $p_{ij} = \mathbb{P}(U_i > U_j)$. The standard Thurstone-Mosteller model assumes $\tilde{\sigma}_i^2 = 1, i = 1, \dots, N$. Mosteller (1951) briefly discussed using a different variance

$\tilde{\sigma}_i^2$ for each player. If we change the link function on p_{ij} in Table 1.2 to the inverse CDF of a standard Normal distribution, Model 1 will coincide with the Thurstone-Mosteller model with player-specific variances when ν is large. However, Mosteller did not discuss this in the context of violation of stochastic transitivity. Our model specification also comes with a purposeful decomposition of the variance σ_i^2 .

If we examine parameters k_i , there are two limiting cases. When $k_i = 0$ for all players, the model reduces to a Bradley-Terry model. If $k_i \rightarrow \infty$ for all players, the model approaches a saturated model with nonidentifiability among the Δ_{ij} and Δ_{ji} .

1.3.1 SETTING HYPERPARAMETERS

Two sets of hyperparameters, (k_α, k_β) and $(\lambda_\alpha, \lambda_\beta)$, need to be determined. To understand how to set the hyperparameter values, we need to understand the role of λ and k_i in the model.

1.3.1.1 RELATIONSHIP BETWEEN λ AND THE DISTRIBUTION OF p_{ij}

Under Model 1 displayed in Table 1.2, λ describes the variability in player strength θ_i . This is reflected in the distribution of p_{ij} for player i . To investigate the effect of λ on distribution of p_{ij} , we fix N , λ and k_0 , and simulate from the prior predictive distribution of Model 1. Figure 1.1 gives the distribution of p_{ij} for $N = 15$ and $k_0 = 0.5$ and 1. Over four different values of λ , we can see that as λ increases, we observe more extreme values in pairwise probabilities. Hence, we can use the understanding of empirical distribution of pairwise probabilities to help us describe likely values of λ .